Presenter guide

Generated by `src/build_deck.js` from the same cues as the speaker notes, so the two cannot drift apart. Talk length: **33.5 minutes** across 34 slides, which leaves room inside a 30 to 40 minute slot.

Time checkpoints (glance at the clock here)

- End of **Motivation** (slide 5): 4.25 min on the clock.
- End of **Data** (slide 11): 11.5 min on the clock.
- End of **Methods** (slide 16): 17 min on the clock.
- End of **Results** (slide 28): 28.5 min on the clock.
- End of **Discussion** (slide 30): 30.25 min on the clock.
- End of **Conclusion** (slide 34): 33.5 min on the clock.

If you are more than two minutes over at the end of Methods, cut slide 12 (model suite), slide 23 (weather through the threshold) and slide 30 (limitations, say two sentences) rather than rushing the results.

How to deliver it

- One message per slide: say the takeaway bar aloud, in your own words, before you leave the slide.
- The talk has one spine: *the alarm budget moves the hit rate by about 50 points, the model by about 5.* Every section either sets it up (data, methods), earns it (RQ1 to RQ3) or defends it (robustness, limits).
- Look at the audience on the headline numbers, and at the slide only to point.
- Never read the notes: they are cues. Rehearse three times aloud with a timer, once with the backup questions B8 and B9.
- Hard questions are welcome: state the limit before they do. The report already does, and the slides say so.

Before the day (yours to do)

- Open the .pptx in PowerPoint (not just a viewer) and flip through once: check line breaks, since this deck was checked with a stand-in for Calibri.
- Send the deck to Dr Tian a week ahead and ask for one round of feedback; he responded quickly to the report.
- Check the oral's marking rubric. If it weights clarity, timing or questions, re-balance: the talk is built to give clarity and a defensible Q&A.
- Rehearse aloud with a timer, three times; once for someone outside mathematics. The test: can they say the main finding back in one sentence (the dial matters more than the detector)?
- Put the PDF on a USB stick and in the cloud as a fallback; check the room's screen ratio (the deck is 16:9).
- Know slides 4, 25 and 28 cold: they carry the idea in one picture, what a budget means, and the central result.

One-page cue card (say the takeaway in your own words, then move on)

Slide	Title	Takeaway
1	Title	The dial matters more than the detector: the alarm budget moves the hit rate by about 50 points, the model by about 5.
2	A forecast is only useful if it arrives before the hazardous day does	A model with good average error can still miss the extreme days that are the only ones anyone cares about.
3	Three research questions, and the third turned out to dominate the other two	The third question turns out to dominate the other two, and the talk is organised around it.
4	The idea in one picture: the detector matters little, the dial matters a lot	Like a smoke alarm: swapping the brand moves the hit rate from 0.82 to 0.87; turning the sensitivity dial moves it from 0.38 to 0.88.
5	The route: from the data, through the models, to the alarm budget	Hold one question in mind: how much does the model matter, compared with the number of alarms you allow?
6	One well-characterised Delhi monitor, nearly ten years of daily PM2.5	One reference site with a full decade of record: good for comparing methods, not a claim about all of Delhi.
7	Two defects in the source data were found and corrected before any model was fitted	Rules: below 0 (5 days), coverage below 50% (270), above 1000 (7): 282 of 3,375 removed, and the exceedance rate moves only 30.0% to 27.0%.
8	Exceedances arrive as multi-day winter episodes, and that is the structure worth forecasting	Episodes make persistence a strong forecast, and the heavy right tail is why every model is fitted to log PM2.5.
9	Every year, the same months: hazardous days gather from October to February	Hazard is seasonal and arrives in episodes, so the calendar alone is already a strong baseline that a model has to beat.
10	An assumption from the approved proposal fails: there is no weekly cycle at this station	Three independent tests agree: ARIMA carries the annual cycle only, with no weekly seasonal order.
11	The annual cycle is unmistakable, so ARIMA carries it with Fourier terms and no weekly order	A simpler, better-identified specification: the data justified dropping the weekly term, and the Fourier terms carry the season.
12	Twelve specifications from three	Fitted to log PM2.5, forecast at h = 1, 2, 3 days, refitted as the training window expands, scored on identical origins by identical

	families, all scored by one protocol	code.
13	Three regressors, each controlling variance by a different mechanism	Trees are fitted to both the level and the change from the origin, because a tree cannot extrapolate and this series drifts.
14	Three networks under one protocol, so the comparison is the cell and nothing else	Only the cell differs: gated recurrence, a memory cell, or attention. Each is reported as a ten-seed range.
15	Rolling-origin evaluation: 2,303 forecasts, none of which can see the future	The protocol is the credibility of every number that follows: leakage-tested, and one draw of history is never trusted.
16	Scored as a warning system: an alarm budget in alarms per year, not just an error	Plain accuracy is never reported: with hazardous days a minority, a model that never alarms scores high by construction.
17	RQ1: the networks win only at one day ahead, and ARIMA beats every ML regressor	Networks: 0.910 to 0.939 vs ARIMA 0.940 at one day. At three days 0.831 vs 0.832 is a tie. Every ML regressor trails ARIMA throughout.
18	Both models follow each winter episode; the networks' edge shows on the sharpest rises	The edge is a short-range one: the forecast tracks the shape of each episode, with the gap on the sharpest rises.
19	A network's result is a range, not a number, and the spread rivals the gaps between models	A single seed could be reported as beating or losing to a rival at the analyst's convenience: only ranges are honest.
20	Formal tests are more conservative than counting seeds, and the picture holds	One-day wins are real but modest; beyond one day nothing is confirmed, and the report says so.
21	Two reasons for ruling models out did not survive being tested	A defensible prior is not evidence: both arguments were tested directly, and neither is supported.
22	Weather helps at one day ahead and leaves two and three days a tie	GRU with weather beats ARIMA at one day in all ten seeds. At two days: 3 win, 6 tie, 1 loses. At three: 3 win, 2 tie, 5 lose.
23	Carried through the decision threshold, a more accurate forecaster buys about one point of hit rate	The ceiling is not the models being too inaccurate: it persists after a materially more accurate forecaster is added.
24	RQ2: forecast-then-threshold has a real but modest edge over direct	Median gap 0.011 of hit rate. Forecast-then-threshold wins 8 of 9

	classification	at a 40-alarm budget, but loses 6 of 9 at 100.
25	What an alarm budget means: of every 100 hazardous days, how many are caught	From one alarm every nine days to one every four, the same model catches 88 of 100 hazardous days instead of 38: a policy choice, not a modelling one.
26	RQ3, step 1: one model, and its hit rate rises as you allow more alarms	Allow more alarms and the same model catches more hazardous days: that is the dial.
27	RQ3, step 2: add three very different models, and the curves nearly coincide	Swap the detector, and the curve barely moves.
28	RQ3: moving the alarm budget changes the hit rate by fifty points, and changing the model by five	The dial matters more than the detector: the alarm budget is the policy instrument, and the model only supplies its input.
29	The central finding survives every check, and the checks changed what I can claim	Bounded checks are single-split, three-seed: nothing in the ten-seed Table 4.1 was changed on their strength.
30	What the results do not show	None of these overturns the central finding: it rests on several independent sources agreeing, not on any single p-value.
31	What this project contributes	Applying a standard model suite to Delhi is not new; evaluating it as a warning system with a varying alarm budget is.
32	Three questions asked, three answered	Machine learning earns a real but narrow place; the dial (the alarm budget), not the detector (the model), decides how many hazardous days are caught.
33	Design the warning system around its alarm budget, and treat the model as a refinement	The recommendation: design a hazardous air quality warning system around the alarm budget its operator can justify, with the forecasting model a secondary refinement.
34	Thank you. Questions?	Thank you.

Timing table

Slide	Title	Minutes	Running total
1	Title	0.5	0.5
2	A forecast is only useful if it arrives before the hazardous day does	1.25	1.75
3	Three research questions, and the third turned out to dominate the other two	1.25	3

Slide	Title	Minutes	Running total
4	The idea in one picture: the detector matters little, the dial matters a lot	0.75	3.75
5	The route: from the data, through the models, to the alarm budget	0.5	4.25
6	One well-characterised Delhi monitor, nearly ten years of daily PM2.5	1.25	5.5
7	Two defects in the source data were found and corrected before any model was fitted	1.5	7
8	Exceedances arrive as multi-day winter episodes, and that is the structure worth forecasting	1	8
9	Every year, the same months: hazardous days gather from October to February	0.5	8.5
10	An assumption from the approved proposal fails: there is no weekly cycle at this station	1.5	10
11	The annual cycle is unmistakable, so ARIMA carries it with Fourier terms and no weekly order	1.5	11.5
12	Twelve specifications from three families, all scored by one protocol	0.75	12.25
13	Three regressors, each controlling variance by a different mechanism	1.25	13.5
14	Three networks under one protocol, so the comparison is the cell and nothing else	1.25	14.75
15	Rolling-origin evaluation: 2,303 forecasts, none of which can see the future	1.25	16
16	Scored as a warning system: an alarm budget in alarms per year, not just an error	1	17
17	RQ1: the networks win only at one day ahead, and ARIMA beats every ML regressor	1.75	18.75
18	Both models follow each winter episode; the networks’ edge shows on the sharpest rises	0.5	19.25
19	A network’s result is a range, not a number, and the spread rivals the gaps between models	1	20.25
20	Formal tests are more conservative than counting seeds, and the picture holds	1	21.25

Slide	Title	Minutes	Running total
21	Two reasons for ruling models out did not survive being tested	1	22.25
22	Weather helps at one day ahead and leaves two and three days a tie	1.25	23.5
23	Carried through the decision threshold, a more accurate forecaster buys about one point of hit rate	0.5	24
24	RQ2: forecast-then-threshold has a real but modest edge over direct classification	1.25	25.25
25	What an alarm budget means: of every 100 hazardous days, how many are caught	1	26.25
26	RQ3, step 1: one model, and its hit rate rises as you allow more alarms	0.5	26.75
27	RQ3, step 2: add three very different models, and the curves nearly coincide	0.5	27.25
28	RQ3: moving the alarm budget changes the hit rate by fifty points, and changing the model by five	1.25	28.5
29	The central finding survives every check, and the checks changed what I can claim	1	29.5
30	What the results do not show	0.75	30.25
31	What this project contributes	1	31.25
32	Three questions asked, three answered	1.25	32.5
33	Design the warning system around its alarm budget, and treat the model as a refinement	1	33.5
34	Thank you. Questions?	0	33.5

Cues, slide by slide

1. Title

- **Say:** Good morning. The answer to build to: the number of alarms a health authority is willing to raise matters many times more than which model produces the forecast.
- **Transition:** First, why the question matters.

2. A forecast is only useful if it arrives before the hazardous day does

- **Say:** Sustained PM2.5 exposure is linked to cardiopulmonary and all-cause mortality, so I forecast a health threshold, not a round number: 121 µg/m³, the CPCB 'Very Poor' boundary. At my monitor 29.7 percent of days exceed it. Existing Delhi machine learning studies report R squared, RMSE and MAE; neither evaluates a warning system. I grew up in India, so this is not an abstract statistic for me.
- **Point to:** the four stat cards, then the takeaway.
- **If asked:** 121 is the CPCB National Air Quality Index boundary of the category below 'Severe', for the 24-hour mean.
- **Transition:** That gap gives me three research questions.

3. Three research questions, and the third turned out to dominate the other two

- **Say:** One: how accurately can concentrations be forecast one to three days ahead by machine learning, against classical and naive benchmarks. Two: forecast the concentration then threshold it, or classify the exceedance directly. Three: as a warning system, what hit rate and false alarm rate does each approach achieve, and how does the decision threshold trade them off. The answers are on the cards; I will earn each one with evidence.
- **Point to:** the bold answer on each card, RQ3 in vermillion.
- **Transition:** First, one picture that holds the whole talk.

4. The idea in one picture: the detector matters little, the dial matters a lot

- **Say:** Think of a smoke alarm. There are two things you can change: which detector you buy, and how sensitive you set the dial. On the left I swap the forecasting model at a fixed budget of 100 alarms a year: hit rates 0.82 to 0.87. On the right I keep the best model and turn the dial from 40 to 100 alarms a year: 0.38 to 0.88. Same axis. The rest of the talk is the evidence for this picture, and the care taken not to fool myself.
- **Point to:** the left panel is almost flat; the right panel climbs. Same vertical scale.
- **If asked:** Both panels are one day ahead, from Table 4.7. The left uses four representative models at 100 alarms (as in the report's Figure 4.5); the weak lag-7 seasonal-naive benchmark would be lower, 0.758. The right uses the best model at each budget.
- **Transition:** Here is the route to that evidence.

5. The route: from the data, through the models, to the alarm budget

- **Say:** Roughly seven minutes on the data and what was wrong with it, five on the models and how they are tested, eleven on results against the three questions, then limits and conclusions. Keep one question in mind: how much does the model matter, compared with the number of alarms you allow?
- **Transition:** Start with the data.

6. One well-characterised Delhi monitor, nearly ten years of daily PM2.5

- **Say:** Data come from OpenAQ. I compared five stations within 25 km on coverage rather than nominal record length and found every CPCB station serves data only from 19 February 2025, about eighteen months, although the metadata advertises 2016. Only location 8118 returns the full record. The cost,

stated plainly: it is an AirNow feed, almost certainly a US diplomatic post monitor, so every number describes one site, not Delhi.

- **Point to:** the table, then the cost box on the right.
- **If asked:** Four independent stations sharing an identical cutoff points to a limit of the measurements endpoint, not coincidence.
- **Transition:** That series had two defects.

7. Two defects in the source data were found and corrected before any model was fitted

- **Say:** First, impossible values: five negatives, and 1990 appearing three times identically, one on 2 September in the monsoon between neighbours of 42 and 178. That is a clipped instrument ceiling. Second, the coverage statistic divided observation count by 24, but the sensor was half-hourly until 2024, so half a day of a polluted afternoon was scored as a complete day. Days at 50 to 75 percent true coverage gave implausible values about seven times as often as days above 90. I took the cadence from the data instead.
- **Point to:** left: sensor cadence against the assumed 24. right: implausible rate by true coverage.
- **If asked:** 1,085 of 3,467 days fall below 75 percent true coverage, far more than the 268 with no value at all. 1000 is a judgement; sensitivity check in the robustness slide.
- **Transition:** With clean data, what does the series look like?

8. Exceedances arrive as multi-day winter episodes, and that is the structure worth forecasting

- **Say:** The cleaned series, with days above the threshold shaded. They come in winter episodes, not isolated days, which is why persistence turns out to be a strong forecast. The distribution is strongly right skewed, skewness 2.17 and excess kurtosis 8.31, so every model is fitted to the logarithm; annual Fourier terms explain 69 percent of the variance of the log.
- **Point to:** the shaded episodes, then the four skew statistics.
- **Transition:** The same data as a calendar.

9. Every year, the same months: hazardous days gather from October to February

- **Say:** Another view of the same data. Each cell is a month; darker means more of its days were above 121. The pattern repeats every year: October to February is dark, June to September is almost clear. This regularity is the annual cycle the models carry, and it is why a calendar-only forecast, climatology, is a surprisingly strong baseline.
- **Point to:** the dark columns, then the pale monsoon months.
- **If asked:** Months with fewer than 15 usable days are left blank. 2016 starts in November and 2026 ends in August.
- **Transition:** Is there any other seasonality, say weekly?

10. An assumption from the approved proposal fails: there is no weekly cycle at this station

- **Say:** The proposal assumed a weekly cycle alongside the annual one and treated two seasonal periods as the central difficulty. I tested it three ways. The periodogram’s dominant periods are the annual cycle and harmonics, nothing near seven days. The autocorrelation at lag 7 sits on the decay curve between lags 6

and 8. Day of week gives a 6.5 percent spread with p = 0.34 and a rank test p = 0.77. Three tests with different assumptions agreeing is what makes the absence convincing.

- **Point to:** 7 days on the periodogram; the circled lags; zero inside every interval.
- **If asked:** A weekly rhythm is a traffic signature; 8118 is in a diplomatic enclave, not roadside like ITO or Anand Vihar.
- **Transition:** So the annual cycle is the only seasonality to model.

11. The annual cycle is unmistakable, so ARIMA carries it with Fourier terms and no weekly order

- **Say:** The annual cycle is the opposite: high in winter, low in the monsoon, which is what makes the missing weekly cycle credible. The benchmark is ARIMA on the log series with four annual Fourier pairs as exogenous regressors. The order was chosen by AIC over a 24-candidate grid: (1,1,1) scores 1892.5, the next, (1,1,2), 1893.6, a gap of 1.1, too small to call, so parsimony decides. Fitted phi 0.581, theta minus 0.987, which is 0.013 inside the invertibility boundary.
- **Point to:** the equation: AR and MA part, then the Fourier sum.
- **If asked:** In the rolling refits theta crosses the boundary at 3 of 28 refits, each with fewer than 1,700 training days; the optimiser is unconstrained so the forecast is still produced and scored. Backup B5.
- **Transition:** Around that benchmark, the model suite.

12. Twelve specifications from three families, all scored by one protocol

- **Say:** Classical and naive: persistence, climatology, seasonal naive at lag 7, and ARIMA. Machine learning: ridge, random forest and histogram gradient boosting, the trees each on the level and on the change from the origin. Deep learning: a GRU, an LSTM and a transformer, each over ten seeds. Dr Tian asked for machine learning rather than relying on ARIMA, and several methods tried; ARIMA stays as the reference point.
- **If asked:** A Kolmogorov-Arnold-LSTM hybrid, which Dr Tian named, was tried as a bounded check only.
- **Transition:** Two lines on the regressors, then the networks.

13. Three regressors, each controlling variance by a different mechanism

- **Say:** Ridge shrinks coefficients and has a closed form: X transpose X plus lambda I is positive definite for any lambda above zero, so a unique solution exists even with collinear lags. A random forest averages B trees; the variance of the average is rho sigma squared plus one minus rho over B sigma squared. The second term vanishes as B grows, the first is a floor, and random feature subsets lower rho. Boosting fits each tree to the pseudo-residual, the negative gradient of the loss, which is descent in function space.
- **Point to:** the three formulas, left to right.
- **If asked:** Trees were fitted to level and change because a tree cannot extrapolate beyond the targets it saw, and this series drifts across the decade. Missing values are never imputed; histogram boosting accepts them natively.
- **Transition:** Now the deep networks.

14. Three networks under one protocol, so the comparison is the cell and nothing else

- **Say:** The GRU has two gates and no separate memory cell; the LSTM three gates and an explicit cell state;

the transformer replaces recurrence with self-attention. All three read the same 30-day window of six channels and predict the change in log concentration, with Adam on absolute error and early stopping, over ten seeds. The LSTM and transformer were added only after Dr Tian's feedback challenged the GRU-over-LSTM and small-data arguments, so I tested them rather than assuming.

- **Point to:** the parameter column: 2,329, 3,097, 3,409, 3,288, confirmed by counting the fitted objects.
- **If asked:** Recurrent nets need a value at every timestep, so gaps are forward-filled with a missingness flag; the transformer reuses that as an implementation choice, not a necessity.
- **Transition:** How do we score them without fooling ourselves?

15. Rolling-origin evaluation: 2,303 forecasts, none of which can see the future

- **Say:** Every model is scored at 2,303 origins over 6.8 years after a three-year burn-in, refitted every 90 days, the networks every 180. The invariant is that at origin t a model may use observations up to and including day t and nothing else. Nothing fails visibly when that breaks; the numbers just improve. So it is tested: corrupt the series from a chosen date, rerun, and require that no earlier forecast changes. A single split would have misled me: ARIMA scored 0.903 on one calm block and 0.940 under the rolling harness.
- **Point to:** the timeline, then 0.903 to 0.940.
- **If asked:** The label on feature row t' is the value at t' + h, so the training set ends at t minus h; hence one model per horizon.
- **Transition:** Now the measures.

16. Scored as a warning system: an alarm budget in alarms per year, not just an error

- **Say:** Accuracy is relative MAE against persistence on identical rows, so 1.000 is persistence. I report it rather than MASE alone because MASE divides by the training period's volatility, so it can fall for every model on a calmer block. The warning measures come from one contingency table: hit rate, false alarm rate, and alarms per year, the unit an operator actually budgets in. I never report plain accuracy, because with hazardous days a minority a model that never alarms scores high.
- **Point to:** the four cards; then the takeaway.
- **If asked:** Diebold-Mariano with the Harvey-Leybourne-Newbold small-sample correction and absolute-error loss, run once per seed.
- **Transition:** Now the results.

17. RQ1: the networks win only at one day ahead, and ARIMA beats every ML regressor

- **Say:** Table 4.1 drawn. Left to right, one, two and three days; rMAE, lower is better, persistence at 1.0. The three networks are bars because each is a range over ten seeds. At one day they run 0.910 to 0.939 against ARIMA at 0.940, so they win. At three days the best network seed, 0.831, and ARIMA, 0.832, are separated by a few parts in ten thousand: a tie. And all five machine learning regressors sit behind ARIMA at every horizon, which fits the M4 competition result that flexible learners do not reliably beat classical benchmarks on a few thousand observations.
- **Point to:** ARIMA diamond against the coloured bars, panel by panel.
- **If asked:** The full Table 4.1 is backup B1.
- **Transition:** What do the forecasts actually look like?

18. Both models follow each winter episode; the networks' edge shows on the sharpest rises

- **Say:** One representative winter, November 2024 to February 2025. Both models track the shape of each episode, not only its level, and the gap between them is visible mainly on the sharpest rises, exactly where a short-range nonlinear model would earn an edge. The GRU line is seed 0 alone, for illustration only; every quantitative claim uses all ten.
- **Transition:** But a single seed can mislead, as the next slide shows.

19. A network's result is a range, not a number, and the spread rivals the gaps between models

- **Say:** This is why I report ranges. The GRU's ten-seed spread is 0.029, 0.028 and 0.048 at the three horizons; at three days that is wider than the gap separating ARIMA from two other models, so a single seed could be reported as beating them or losing to them at the analyst's convenience. Even tree models moved by 0.004 between two machines. And rerunning the same seed on the same machine moved the GRU almost as much as a new seed, so I doubled the sample from five to ten.
- **Point to:** the dashed floor of 0.004 against the bars.
- **If asked:** Thread-order floating point in the OpenMP-parallel gradient boosting explains the 0.004; for networks, training accumulates small floating-point differences.
- **Transition:** Then are the one-day wins statistically real?

20. Formal tests are more conservative than counting seeds, and the picture holds

- **Say:** A Diebold-Mariano test against ARIMA for each seed. At one day the statistic favours the network in all ten seeds for the GRU but is significant in only four; five for the LSTM, seven for the transformer. At two and three days no network reaches significance, bar one GRU seed at three days, in ARIMA's favour. For the five ML regressors, all fifteen comparisons favour ARIMA and fourteen reach p below 0.05. A limitation I want to state myself: twenty-seven tests with no correction; after Bonferroni only six survive and none of the plain GRU, LSTM or transformer, so read beyond one day as suggestive.
- **Point to:** the shaded one-day column; then 6 of 27.
- **If asked:** Wilson 95% interval for 4 of 10 is 0.17 to 0.69; bootstrapping the statistic for heavy tails was not done.
- **Transition:** Now two arguments I made and then tested.

21. Two reasons for ruling models out did not survive being tested

- **Say:** In the literature review I argued for the GRU on two grounds: fewer parameters, and too little data for a transformer. I tested both under the identical protocol, on Dr Tian's feedback. The LSTM beats the GRU at every horizon, by 0.005, 0.008 and 0.013, despite a third more parameters. Dr Tian said the series, counting every evaluation block, is some four thousand days, which clears the bar for a transformer. The ten-seed run bears him out: the transformer beats the GRU in eight of ten matched seeds at every horizon and has the strongest formal significance at one day. One caveat: its four settings were inherited, not tuned.
- **Transition:** Next: does adding weather change the answer?

22. Weather helps at one day ahead and leaves two and three days a tie

- **Say:** I added observed wind speed and temperature; a calm, stagnant air mass is the physical candidate that the calendar-only Fourier terms cannot represent. At one day the weather network separates cleanly from the plain one; at two and three the ranges overlap. Gradient boosting improves everywhere, the random forest gets worse at three days, so weather is not a free improvement. Important: under the first five seeds two and three days both looked like wins. Doubling the seeds overturned that.
- **Point to:** blue bars (with weather) against orange (plain), relative to the ARIMA line.
- **If asked:** The weather record ends 202 days before the PM2.5 series; I repeated the comparison inside the fully observed window and the picture held.
- **Transition:** Does that accuracy gain matter once it meets the alarm budget?

23. Carried through the decision threshold, a more accurate forecaster buys about one point of hit rate

- **Say:** The test that matters. I reran the threshold sweep with the weather models as extra candidates. Where they win, the gain is at most 1.2 points of hit rate; elsewhere they are not even the best option. Compare that with the fifty points the budget buys. A forecast that is 8 points more accurate in rMAE buys at most a little over a point.
- **Point to:** the shaded cells: the largest gains.
- **Transition:** Now the second research question.

24. RQ2: forecast-then-threshold has a real but modest edge over direct classification

- **Say:** To compare fairly I matched each classifier to the regressor built on the same algorithm, so the training objective is the only thing that varies. Across three pairs, three horizons and four budgets, 36 comparisons: forecast-then-threshold wins 21, classification 11, four ties. The median gap is 0.011 of hit rate, the largest 0.056. The edge shrinks as the budget loosens: 8 of 9 at a budget of 40, but classification wins 6 of 9 at 100. I prefer forecast-then-threshold also because it yields a concentration that can be rethresholded for any future health policy without refitting.
- **Point to:** shaded cells: the better arm in each column.
- **If asked:** All three horizons are backup B6.
- **Transition:** Now the central result, RQ3, starting with what a budget means.

25. What an alarm budget means: of every 100 hazardous days, how many are caught

- **Say:** Translate the dial into something you can feel. Each grid is 100 hazardous days; orange dots are the ones the best model caught at one day ahead. At 40 alarms a year, one alarm about every nine days, it catches 38 of 100. At 100 alarms, one about every four days, it catches 88. Nothing changed except the dial. This is a policy choice about how many alarms the public will tolerate, not a modelling choice.
- **Point to:** left to right, the orange fills up.
- **If asked:** Hit rates are Table 4.7, best model at one day ahead (0.379, 0.567, 0.735, 0.881), rounded to whole days out of 100. Alarms every N days is 365.25 divided by the budget.
- **Transition:** Now let me show how this comes out of the model comparison, step by step.

26. RQ3, step 1: one model, and its hit rate rises as you allow more alarms

- **Say:** Now the chart the project is built around. Horizontal: alarms raised per year, the dial. Vertical: the share of hazardous days caught. This is one model, ARIMA. As the budget rises, so does the hit rate. That is true by construction, so on its own it proves nothing about models. The question is what happens when I change the model.
- **Point to:** the two axes, then the single navy curve.
- **Transition:** Add three very different models.

27. RQ3, step 2: add three very different models, and the curves nearly coincide

- **Say:** Now add gradient boosting, persistence, which just repeats yesterday, and climatology, which uses only the calendar. Four very different forecasters, one almost identical curve. I have not separated them for legibility, because the overlap is the finding.
- **Point to:** how closely the four lines lie on top of one another.
- **Transition:** Now quantify the two effects.

28. RQ3: moving the alarm budget changes the hit rate by fifty points, and changing the model by five

- **Say:** Now the two effects. Along the dial, from 40 to 100 alarms a year, the best hit rate goes from 0.379 to 0.881: fifty points. At a fixed 100 alarms, switching the model moves it from about 0.82 to 0.88: about five. At three days and 40 alarms a year, persistence, which just repeats yesterday, is the best model in the whole study, 0.359. At 100 alarms a climatological forecast comes within a hundredth of the best, 0.816 against 0.827.
- **Point to:** the vermillion arrow on the left, then the blue bracket at 100 alarms.
- **If asked:** A bigger budget raising the hit rate is true by construction, which is exactly why every model comparison is made at a fixed budget. The finding is the size of the two effects, not that the first exists. How big is the model effect exactly? The report says about five points; recomputed from the sweep it averages 7 across all ten models (2 to 12 by horizon and budget) and 4 without the lag-7 benchmark. Full Table 4.7 is backup B7.
- **Transition:** Why this happens: episodes. Then what I checked, and what I cannot claim.

29. The central finding survives every check, and the checks changed what I can claim

- **Say:** The 1000 cleaning ceiling is a judgement, so I restored the seven removed days and refitted everything. ARIMA barely moves, 0.935 against 0.940, because logs and differencing absorb the contamination. The ML models are not protected: ridge goes from a near tie with persistence to a clear loss, 0.997 to 1.118. So RQ1 strengthens. The bounded checks on the GRU show its three-day configuration is the one I can least defend: a 45-day window, four Fourier harmonics, and SGD each improve it by two to four points. I did not rerun the full evaluation with those, so Table 4.1 is unchanged.
- **Point to:** the first and last rows.
- **Transition:** And the limits.

30. What the results do not show

- **Say:** Limits, stated plainly. One monitor in one city, a diplomatic enclave. Observed weather, not forecast weather, so an operational system might do better. The LSTM and transformer were not given weather or the threshold sweep. Tree grids were small, so ML results are a lower bound. The significance tests are uncorrected with best-seed selection. And network training is not bit-reproducible even at a fixed seed. None of these overturns the central finding.
- **If asked:** Why none overturns it: the finding rests on several independent sources agreeing, not on one p-value.
- **Transition:** So what does the project contribute?

31. What this project contributes

- **Say:** Four things. A warning-system evaluation with an explicit alarm budget, which the Delhi studies I found do not do. A leakage-tested rolling-origin comparison from classical to deep models, with the networks reported as ten-seed ranges. Two defects in the source monitor diagnosed and corrected. And three priors tested instead of assumed: no weekly cycle, the GRU over the LSTM, and the small-data argument against a transformer.
- **Transition:** Back to the three questions.

32. Three questions asked, three answered

- **Say:** RQ1: ARIMA beats every ML regressor at every horizon, with a Diebold-Mariano test at each comparison; the networks beat it only at one day ahead, as ranges across ten seeds. RQ2: neither arm dominates, 21, 11 and 4; I prefer forecast-then-threshold because a concentration can be rethresholded. RQ3: the budget moves hit rate by fifty points, the model by about five. Dr Tian's instruction to try several methods rather than default to ARIMA was right: it is what makes the negative result at two and three days credible, and what surfaces the one-day result.
- **Transition:** What follows from that.

33. Design the warning system around its alarm budget, and treat the model as a refinement

- **Say:** The recommendation: design a hazardous air quality warning system around the alarm budget its operator can justify, with the forecasting model a secondary refinement. Six extensions follow from the report: a real numerical weather forecast; boundary layer height from a reanalysis such as ERA5; the GRU with four Fourier harmonics under the full ten-seed evaluation; a second, independent station; completing the LSTM and transformer with weather and the threshold sweep; and a full ten-seed evaluation of the Kolmogorov-Arnold-LSTM. Thank you; I am happy to take questions.
- **Transition:** Questions.

34. Thank you. Questions?

- **Say:** Thank you. Happy to take questions. Detailed tables are in backup B1 to B7; the questions to expect are B8 and B9.

B1. B1. Report Table 4.1 in full: rMAE against persistence

- **Say:** Full accuracy table. Networks are ranges over ten seeds; everything else is a single fit. Bias is at h = 1 and its units are µg/m³.

B2. B2. The ARIMA order search in full: all 24 candidates by AIC

- **Say:** Table 3.4. Four orders did not converge to a stable estimate on the training block and are marked. (1,1,1) is chosen; the next best is 1.1 AIC points behind.

B3. B3. Diebold-Mariano tests against ARIMA

- **Say:** Tables 4.2 and 4.3. Negative statistic favours the network; for the regressor table a positive statistic favours ARIMA. All 15 regressor pairings favour ARIMA and 14 reach p below 0.05.

B4. B4. GRU sensitivity: three bounded checks, three seeds, one split

- **Say:** Table 5.1 and Figure 5.1. Mean rMAE across the same three seeds. At one and two days the default sits near the best; at three days it is the worst of six architecture settings. Single split, three seeds: nothing in Table 4.1 changed on this basis.

B5. B5. ARIMA coefficient stability, and all four architectures' bounded checks

- **Say:** Figure 3.6: phi drifts upward; theta oscillates near the invertibility boundary and crosses it at three of 28 refits, each with fewer than 1,700 training days. The optimiser is unconstrained, so a forecast is still produced there. Figure 5.2: all four architectures' bounded checks; the KAN hybrid sits among the other three.

B6. B6. RQ2 at all three horizons: the hit-rate advantage of forecast-then-threshold

- **Say:** Figure 4.4 extends Table 4.6 to all horizons. Above zero, forecast-then-threshold wins. It wins 8 of 12 matched comparisons at one day, 7 of 12 at two, 6 of 12 at three: an edge that narrows rather than changing hands.

B7. B7. Report Table 4.7 in full: best hit rate within an alarm budget

- **Say:** The principal table. Persistence and climatology at h = 3 are the two models the discussion names, so those claims can be checked against a printed cell.

B8. B8. Likely questions, 1 of 2: scope and framing

- **Say:** Cue sheet for questions. Answers are short on purpose; the evidence is in the report section named.

B9. B9. Likely questions, 2 of 2: evidence and method

- **Say:** Cue sheet for questions, continued.

B10. B10. The words used in this talk, in plain language

- **Say:** Glossary. Useful if an audience member is outside statistics: every term the talk uses, in one line each.